

POLYMER PRODUCTS

QUALITY ASSURANCE PLAN-ELASTOMERIC BEARING

Table - 1(Tests on Raw Material)

SL No.	Component to be Tested	Test to be conducted	No. Of samples	Reference Documents	Acceptance Criteria	Testing Agency	Remarks
.	Rubber Compound (Used in manufacturing of Elastomeric Bearing)	1. PHYSICAL PROPERTIES 1.1 Hardness 1.2 Min. Tensile strength 1.3 Min. elongation at break 2. Max. Compression set 3. Accelerated ageing 3.1 Maximum Change in Hardness 3.2 Maximum Change in Tensile strength 3.3 Maximum Change in Elongation 4. Adhesion Strength 5. Specific Gravity 6. Ash Content 7. Ozone Resistance Test	1 Dumbels for 3 order 2 Buttons for 3 order 2/3 Dumbels for 3 order One Sample for 3 order - Do - - Do - - Do -	IS:3400 (Part II) ISO - 37 IS:3400 (Part I) ISO 815 ISO – 48 ISO - 188 IS:3400 (Part XIV) IRC : 83 (Part II)-2018 CL: 7.9.1.2 IRC : 83 (Part II)-2018 CL: 7.9.1.2 ISO 1431-1	IRHD 60 $\pm$ 5 16 N/mm ² 425 % 15% max. $\pm$ 5 $\pm$ 15 $\pm$ 25 Min 7 kN/m $\pm$ 0.2 3.50 % To 4.50% No Crack	Manufacturer's Laboratory -do- -do- -do- -do- -do- - - 3 rd Party (Independent Lab)	Client/TPI Client/TPI Client/TPI Review ITC Review ITC Review ITC Review ITC Client/TPI Review ITC Review of 3 rd party lab test certificate Review of 3 rd party lab test certificate

2.	M. S. Plate (Reinforced Plate)	▪ <u>PHYSICAL TESTS:</u> 1. Ultimate Tensile strength 2. % Elongation ▪ <u>CHEMICAL TESTS</u> - C, Mn, S, Si, and P	One piece per lot of procurement	IS:2062	Test Certificate	NABL approved Laboratory	Review of 3 rd party lab test certificate
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Table - 2 (Finished Bearings)

SL No.	Component to be Tested	Test to be conducted	No. Of samples	Reference Documents	Acceptance Criteria	Testing Agency	Remarks
1.	Finished Bearing	1) Visual Inspection	10% of Lot (random selection)	IRC : 83 (Part II)-2018	No Visual defect shall be observed.	In House	Client/TPI
		2) Dimensional Inspection	10% of Lot (random selection)	IRC : 83 (Part II)-2018	IRC : 83 (Part II)-2018	-do-	Client/TPI
		3) Hardness Test	10% of Lot (random selection)	IRC : 83 (Part II)-2018	IRC : 83 (Part II)-2018	-do-	Client/TPI
		4) Determination of Shear Modulus (G)	2 Nos. per Lot	IRC : 83 (Part II)-2018	IRC : 83 (Part II)-2018	-do-	Client/TPI
		5) Maximum Compressive Stiffness	1 Nos per Lot	IRC : 83 (Part II)-2018	IRC : 83 (Part II)-2018	-do-	Client/TPI
		6) Shear Bond Test	2 Nos per Lot	IRC : 83 (Part II)-2018	IRC : 83 (Part II)-2018	-do-	Client/TPI

Manufacturer

Polymer Products